

WASP-43b

R-0644

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_170328 · created 2026-07-25T19:48:37+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457840.7624 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.003 +/- 0.059
Transit depth (Rp/Rs)^2	0.00073 +/- 0.032 [%]
Orbital Inclination (inc)	87.09 +/- 2.88
Airmass coefficient 1 (a1)	0.28 +/- 0.13
Airmass coefficient 2 (a2)	0.8 +/- 0.29
Scatter in the residuals of the lightcurve fit is	45.57 %
Best Comparison Star	None
Optimal Aperture	1.03
Optimal Annulus	3.87
Transit Duration (day)	0.0526 +/- 0.0049

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.003 $\pm$ 0.059 vs 0.1594 (deviation 2.65 σ)
Duration fit vs published	0.0526 d vs 0.0512 d (deviation 0.28 σ)
Mid-time O-C	0.6 min
Depth SNR	0.1
Residual scatter	45.57 %

Light curve

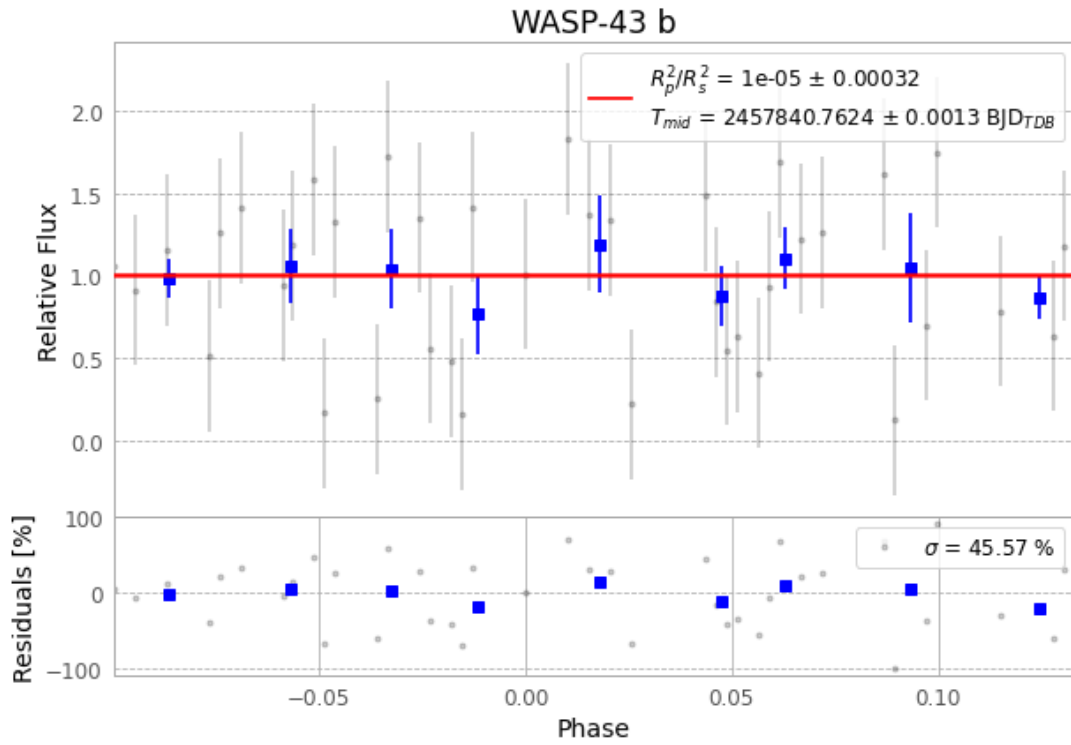


Figure 1 — FinalLightCurve_WASP-43 b_2017-03-27.png (SHA-256 6d80870b8f3e3b88...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [503, 153], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9cdd2799f098040f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

92 FITS frames (60 MB), WASP-43170328041212.FITS → WASP-43170328084512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-170328.log (f05bca4da11c...), FinalLightCurve_WASP-43 b_2017-03-27.png (6d80870b8f3e...), FinalLightCurve_WASP-43 b_2017-03-27.csv (7520c092ae16...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 19:48Z.